Answers · Term 1

TERM 1

Teacher copy. Answers were calculated in code and checked. Circumference and Pythagoras answers use the π and √ keys; where π ≈ 3.14 gives a different 1 d.p. answer it is listed as an alternative. Reasoning answers are model responses: accept any correct explanation.

WK 1	Q1	Q2	Q3	Q4	Q5
S1	$x = 51$	-6	$k = 14$	6	30 m^2
S2	45 m^2	$2x$	9 stalls	$x = 51$	$\frac{2}{3}$
S3	80 km	$x = 71$ (vertically opposite angles are equal)	14	157.1 cm Also accept 157.0 cm if π ≈ 3.14 is used	0.85

WK 2	Q1	Q2	Q3	Q4	Q5
S1	75%	1	24 cm^3	$x = 65$	No: substituting $a = 11$ does not make both sides equal (the solution is $a = 10$).
S2	$\frac{4}{5}$ (0.8)	20 m	$x = 139$	0.00519	$30x$
S3	$m = 14$	19 cm	$x = 58$	106 cm^2	40%

WK 3	Q1	Q2	Q3	Q4	Q5
S1	$x = 130$	27	$x = 0$: $y = 9$, $x = 2$: $y = 7$	6	40 mm^2
S2	100 m^3	Sam added first; × must be done before +. Correct answer 27.	$x = 116$	13	$10a + b$
S3	$x = 125$	13	50 000 cm^2	18	$2m + 3$

WK 4	Q1	Q2	Q3	Q4	Q5
S1	7.5	1500 cm^3	$x = 58$	(2, −3)	$\frac{3}{4}$
S2	50 cm^2	$\frac{3}{4}$	$x = 52$	$4y$	7
S3	$k = 20$	The square: rectangle 44 m^2, square 49 m^2.	8.1	20%	$x = 106$

WK 5	Q1	Q2	Q3	Q4	Q5
S1	The mean (8) is larger than the median (6) because the outlier 19 pulls the mean up.	160 m^2	$x = 60$	$2 \times 7 \times 7$	$p = 6$
S2	37.7 cm	$x = 73$	$0.8\dot{3}$	51	0
S3	136 L	$\frac{1}{10}$ (0.1)	−2	$x = 123$	180 cm^3

Answers · Term 1

TERM 1

WK 6	Q1	Q2	Q3	Q4	Q5
S1	84 mL	14	42 cm	20 students	$x = 63$
S2	40% profit	$x = 40$ (complementary angles add to 90°)	3	27 m	196 cm^2
S3	80	$x = 130$	84 cm^2	0.025	8

WK 7	Q1	Q2	Q3	Q4	Q5
S1	9^5	56%	$x = 112$	$\frac{1}{7}$	$y = 6$
S2	$187	5^6	42	72 m^3	$x = 71$
S3	The base stays 5; only the indices are added: $5^3 \times 5^5 = 5^8$, not 25^8.	$x = 92$	4	30 m^2	$10m$

WK 8	Q1	Q2	Q3	Q4	Q5
S1	$2^3 \times 3^2$	28 m^2	$\frac{3}{5}$	No: when $x = 4$, $y = 18$, not 19.	$x = 132$
S2	82	40%	10 000	$x = 92$	$x = 0$: $y = 6$, $x = 2$: $y = 4$
S3	5^4	$7p$	$\frac{3}{4}$	72 cm^3	$x = 126$

WK 9	Q1	Q2	Q3	Q4	Q5
S1	9^9	3	12	50.3 m Also accept 50.2 m if π ≈ 3.14 is used	$4x$
S2	$345	5^4	$\frac{13}{60}$ (0.217)	Sam added first; × must be done before +. Correct answer 115.	$x = 115$
S3	2×3^3	7 °C	1.25 L	15	$x = 100$

WK 10	Q1	Q2	Q3	Q4	Q5
S1	9^5	6	74 cm	$x = 31$	−10
S2	6^{16}	30 km	$\frac{4}{13}$	65 cm^2	0.35
S3	Yes: $12^2 + 16^2 = 400 = 20^2$	$\frac{8}{9}$	62	$x - 5$	$x = 65$